

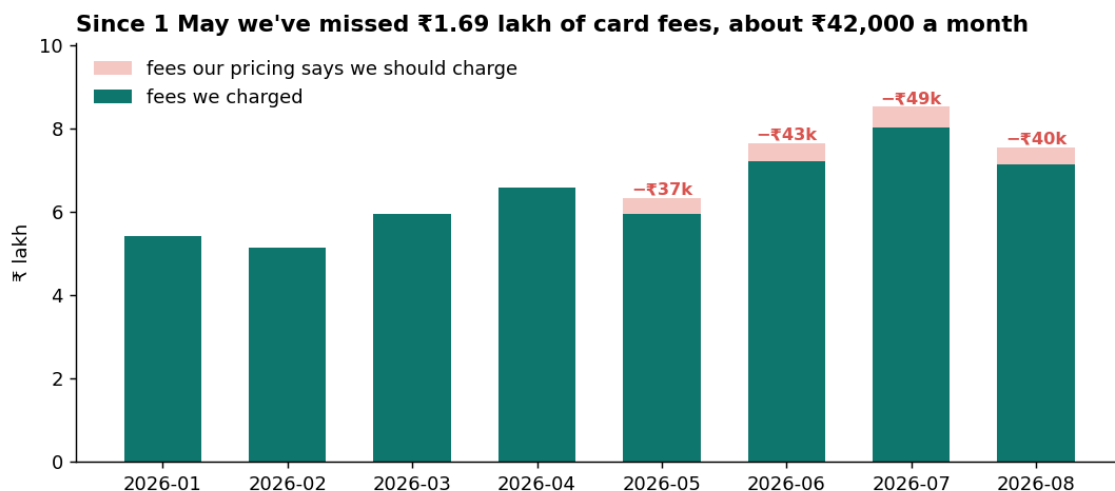
Tarang Pay · where should October's one sprint go?

Memo to the Head of Payments Operations · data 1 Jan – 31 Aug 2026 · ₹101 crore of TPV · fictional company and data

Business problem: payment volume nearly doubled this year (₹8.9 crore to ₹17 crore a month), but fee revenue grew only by a third. Numbers that didn't match between systems, a refund spike and late payouts left leadership unsure which figures to trust before the investor update. **Main question:** which one fix should get October's engineering sprint?

Recommendation

- **Spend the October sprint on option B: fix the Growth-plan card pricing and recover the missed fees.** Since 1 May, 85 merchants have paid ₹0 on card payments. That's ₹1.69 lakh so far and about ₹42,000 a month (5.9% of fee revenue). Ask sales first whether any were promised 0%.
- **Escalate Nilgiri Bank now, and make option A the next sprint.** Since May, about 26% of its payments reach merchants a working day late, against about 3% at other banks: about ₹45 lakh of payouts a month, across 773 merchants, many of them our largest.
- **Fix the monthly report before the investor update.** Report success by order (93.3%, not 88.8%); add the 184 payments paid out while marked failed to July TPV (₹4.77 lakh); restore June's missing ledger entries (₹74,910); show refunds by the month of the original payment.
- **Ask engineering about 1-24 July:** 184 payments timed out on our side but were captured by the bank. Check whether those customers paid twice.
- **Watch new merchants' checkouts.** Merchants with technical errors on more than 4% of first-month payments were still active three months later 70% of the time, against 91% for the rest.



What the chart says: fix the Growth-plan card pricing now: every month it waits costs about ₹42,000.

Headline number: **₹1.69 lakh of card fees not charged since 1 May.** It's in rupees, exact, has one cause and one owner, and grows every month. For the investor deck I'd use **net take rate, shown next to UPI share:** TPV rewards volume that earns nothing, and success rate doesn't connect to money.

Evidence

- **Take rate fell from 0.61% to 0.44%, mostly for a good reason:** UPI, which earns 0%, grew from 57% to 67% of TPV. The rate on card, netbanking and wallet payments also dropped in May, and about three-quarters of that drop is the ₹0 card fees.
- **On-time payouts fell from 97.4% to 94.3% in May.** Nilgiri Bank started sending success statuses 3–30 hours late, so many miss the 23:00 cut-off. The other late payouts (11,253 across eight months) are the cut-off working as designed.
- **Every successful payment that was never paid out is explained:** most weren't due yet when the data ends, and 738 (₹13.44 lakh) belong to 26 merchants on KYC hold, which the rules allow.
- **Failures:** 61% are the customer's choice. Konkan Bank netbanking fails 40% of the time from 7 to 11 pm, against 18% at other banks.
- **The August refund spike is not an error:** 1,663 refunds (₹18.60 lakh) in one week are returns from the 7–10 August sale.

Caveats

The company and data are fictional. "Potential leakage" stays potential until sales confirms what the 85 merchants were promised. The retention link rests on 109 merchants, and a failing checkout may signal a weaker business rather than cause merchants to leave. August refunds are incomplete: refunds can arrive up to 90 days later.

Appendix · Investigation log

What I noticed	Rows and ₹	What I think happened	Handling	Why
Paid out, but recorded as failed	184 payments · ₹476,792	A timeout change on our side (1–24 Jul) marked slow bank approvals as failed; the bank captured the money	Flagged but kept; counted as successful	Money moved and the merchant was paid; the source record is engineering's to fix
Card fees of ₹0 for Growth-plan merchants	3,937 payments · ₹168,853 not charged	Billing set card rates to 0% for part of the 1 May Growth-plan migration	Left alone; reported as potential leakage	Fees charged are the fact; whether they were promised is for sales and finance to confirm
Settlement batches missing from the ledger	1,177 batches · ₹74,910 of fees	Finance's daily booking job didn't run on 16–18 June	Flagged	Finance owns the books; the correct amount is known
Refund spike, week of 24 Aug	1,663 refunds · ₹1,860,297	Returns from the 7–10 Aug sale processed together	Left alone	Every refund is valid; it's a timing effect
Successful payments never paid out	see Task 3	38 merchants on KYC hold	Left alone	Correct behaviour under the rules
97 pending attempts	97 attempts	Started on 30–31 Aug; statuses not in yet	Excluded from rates	Within the 48-hour window
Payouts a working day late (Nilgiri Bank)	7,781 payments since May	Nilgiri sends success statuses 3–30 h late, missing the 23:00 cut-off	Left alone; escalated	The rule worked; the bank feed is the problem
Payouts a working day late (other)	11,253 payments	Payment or status after 23:00	Left alone	Expected under the cut-off rule
Monthly report: success rate	every month	Counts attempts, so retries look like lost payments	Recalculated by order	Orders are what merchants get paid for

Questions I would ask, and what I assumed meanwhile

Question	Assumption I made
Did payment timeout handling change on 1 July and roll back on 24 July? Were any customers charged twice?	The bank captured these payments, so they count as successful
Were any Growth-plan merchants promised 0% on cards?	No: the pricing table is the agreement, so the gap is potential leakage
Did the ledger booking job fail on 16–18 June?	Yes: I used settlement fees for revenue and reported the ledger gap separately
Does Nilgiri Bank know about a status delay since May?	It's on the bank's side: it affects one bank and starts on one date
Can refunds fail or be reversed?	No: the refunds file shows completed refunds only
What counts as an active merchant?	At least one successful payment in the month (or 30-day window)

Working: notebooks/tarang_pay_analysis.ipynb reruns from the data files. How AI was used: AI_USAGE.md.